

Cross-section-averaged oxygen transport along an oxidized grain boundary: lateral closure from the separable solution of the Laplace equation

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Purpose and result

The GB oxidation model reduces oxygen transport to one dimension along the boundary. This note treats the entire oxide cross-section as the transport channel and derives, from the two-dimensional diffusion equation,

- (i) the one-dimensional balance for the cross-section-averaged oxygen concentration $\bar{C}_O$;
- (ii) a closure that gives the interfacial flux J_r directly from $\bar{C}_O$, the oxide half-width L , the diffusivity D_O^{eff} and the interfacial reaction law $J_r(C_O^{\text{int}})$.

The closure is obtained from the separable solution of the Laplace equation in the oxide cross-section. It reduces to one scalar equation for a dimensionless eigenvalue $\mu \in (0, \pi/2)$,

$$\boxed{\frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L} = \sum_k J_r^k(\bar{C}_O \mu \cot \mu)} \quad (1)$$

after which

$$\boxed{J_r = \frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L}, \quad C_O^{\text{int}} = \bar{C}_O \mu \cot \mu} \quad (2)$$

Equation (1) has a unique root and is solved pointwise by Newton's method, exactly as same as the present quasi-steady closure solved for C_O^{int} .

1 Geometry and notation

- y : coordinate along the grain boundary, $y = 0$ at the surface.
- x : coordinate normal to the boundary, $x = 0$ on the boundary mid-plane (plane of symmetry). Only the half-domain $0 \leq x \leq L$ is considered.

- $L(y, t)$: distance from the mid-plane to the oxide/metal interface. After the original boundary channel is filled.
- $C_O(x, y, t)$: concentration of mobile (dissolved) oxygen in the oxide.
- $D_O^{\text{eff}}(y, t)$: effective oxygen diffusivity of the locally mixed oxide, Eq. (eq:Deff) of `main.tex`. It is uniform across the section, because the phases are mixed locally rather than layered.
- Cross-section average, mid-plane value and interface value:

$$\bar{C}_O(y, t) = \frac{1}{L} \int_0^L C_O dx, \quad C_O^c = C_O(0, y, t), \quad C_O^{\text{int}} = C_O(L, y, t). \quad (3)$$

- Interfacial oxygen flux (per unit boundary area, positive toward the metal):

$$J_r(y, t) = -D_O^{\text{eff}} \left. \frac{\partial C_O}{\partial x} \right|_{x=L}. \quad (4)$$

- Interfacial reaction law: the total consumption $J_r(C_O^{\text{int}}) = \sum_k J_r^k(C_O^{\text{int}})$ of Eqs. (eq:rate-cr2o3)–(eq:rate-total) of `main.tex`. It is a nondecreasing function of C_O^{int} at fixed metal concentrations. Where a linearized form is used below, $k \equiv dJ_r/dC_O^{\text{int}}$.

2 Assumptions

- A1. Two-dimensional Fickian diffusion of oxygen in the oxide with a single, isotropic D_O^{eff} that does not vary across the section.
- A2. Symmetry at the mid-plane: $\partial C_O / \partial x = 0$ at $x = 0$.
- A3. Slow variation along the boundary of the half-width, the diffusivity and the metal concentrations that enter the reaction law: $|\partial L / \partial y| \ll 1$, and D_O^{eff} , $J_r(\cdot)$ change little over a distance L .
- A4. Lateral quasi-steady state: the lateral profile adjusts on the time scale L^2 / D_O^{eff} , which is short compared with the time scales on which $\bar{C}_O$ and L change.
- A5. Locally, the lateral profile is the fundamental separable mode of the Laplace equation (Sec. 4). Higher modes decay over a distance of order $0.3L$ or less and are neglected (Sec. 6).

No assumption is made on whether the interface is reaction- or diffusion-limited, and no assumption is made that C_O is uniform across the section.

3 Cross-section integration of the 2D balance

The two-dimensional conservation equation in the oxide is

$$\frac{\partial C_O}{\partial t} = \frac{\partial}{\partial y} \left(D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \right) + D_O^{\text{eff}} \frac{\partial^2 C_O}{\partial x^2}. \quad (5)$$

Integrate each term over $0 \leq x \leq L(y, t)$, using the Leibniz rule where the limit depends on y or t .

Lateral term. With A2 and the definition (4),

$$\int_0^L D_O^{\text{eff}} \frac{\partial^2 C_O}{\partial x^2} dx = D_O^{\text{eff}} \frac{\partial C_O}{\partial x} \Big|_0^L = -J_r. \quad (6)$$

Time term. Because the upper limit moves, the Leibniz rule gives the identity

$$\frac{\partial}{\partial t} \int_0^L C_O dx = \int_0^L \frac{\partial C_O}{\partial t} dx + C_O(L) \frac{\partial L}{\partial t}, \quad \text{i.e.} \quad \int_0^L \frac{\partial C_O}{\partial t} dx = \frac{\partial}{\partial t} (L \bar{C}_O) - C_O^{\text{int}} \frac{\partial L}{\partial t}. \quad (7)$$

The last term is the dissolved oxygen, at concentration C_O^{int} , contained in the oxide formed during dt .

Along-boundary term. Applying the Leibniz rule twice,

$$\begin{aligned} \int_0^L \frac{\partial}{\partial y} \left(D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \right) dx &= \frac{\partial}{\partial y} \int_0^L D_O^{\text{eff}} \frac{\partial C_O}{\partial y} dx - D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \Big|_{x=L} \frac{\partial L}{\partial y}, \\ \int_0^L D_O^{\text{eff}} \frac{\partial C_O}{\partial y} dx &= D_O^{\text{eff}} \left[\frac{\partial}{\partial y} (L \bar{C}_O) - C_O^{\text{int}} \frac{\partial L}{\partial y} \right] = D_O^{\text{eff}} L \frac{\partial \bar{C}_O}{\partial y} + D_O^{\text{eff}} (\bar{C}_O - C_O^{\text{int}}) \frac{\partial L}{\partial y}. \end{aligned} \quad (8)$$

Hence

$$\int_0^L \frac{\partial}{\partial y} \left(D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \right) dx = \frac{\partial}{\partial y} \left(D_O^{\text{eff}} L \frac{\partial \bar{C}_O}{\partial y} \right) + \underbrace{\frac{\partial}{\partial y} \left[D_O^{\text{eff}} (\bar{C}_O - C_O^{\text{int}}) \frac{\partial L}{\partial y} \right]}_{O(\partial L / \partial y)} - D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \Big|_{x=L} \frac{\partial L}{\partial y}. \quad (10)$$

Result. Collecting (6)–(10),

$$\frac{\partial}{\partial t} (L \bar{C}_O) = \frac{\partial}{\partial y} \left(D_O^{\text{eff}} L \frac{\partial \bar{C}_O}{\partial y} \right) - J_r + C_O^{\text{int}} \frac{\partial L}{\partial t} + O\left(\frac{\partial L}{\partial y}\right). \quad (11)$$

Equation (11) is exact. Under A3 the last group is dropped.

At the moving interface the diffusive flux J_r of (4) supplies both the reaction and the dissolved oxygen in the fresh oxide (Stefan condition, dissolved oxygen in the metal neglected):

$$J_r = \sum_k J_r^k(C_O^{\text{int}}) + C_O^{\text{int}} \frac{\partial L}{\partial t}. \quad (12)$$

Substituting (12) into (11), the terms $C_O^{\text{int}} \partial L / \partial t$ cancel and the sink is the reaction consumption alone. The working one-dimensional equation is

$$\boxed{\frac{\partial}{\partial t}(L \bar{C}_O) = \frac{\partial}{\partial y} \left(D_O^{\text{eff}} L \frac{\partial \bar{C}_O}{\partial y} \right) - \sum_k J_r^k} \quad (13)$$

With $\partial L / \partial t = \Omega \sum_k J_r^k$ and $\Omega = 1/C_O^{\text{oxide}}$ the oxide volume formed per unit oxygen consumed, the ratio of the Stefan term in (12) to the reaction term is $C_O^{\text{int}}/C_O^{\text{oxide}} \ll 1$. The diffusive flux J_r and the reaction consumption $\sum_k J_r^k$ therefore differ by a negligible amount, and from here on the single symbol J_r is used for both; in particular the closure of Sec. 5 equates J_r of (4) to $\sum_k J_r^k$. Two remarks:

- The along-boundary conductance is $D_O^{\text{eff}} L$ with the *same* D_O^{eff} as in the lateral direction. This follows from A1 alone; no assumption on the lateral profile was used.
- L sits inside the time derivative. Expanding, $L \partial \bar{C}_O / \partial t = \partial_y(D_O^{\text{eff}} L \partial_y \bar{C}_O) - J_r - \bar{C}_O \partial L / \partial t$; the last term is the dilution of dissolved oxygen as the section widens.

Equation (13) contains two unknowns per position, $\bar{C}_O$ and J_r . The relation between them is derived next.

4 Lateral problem: separation of variables

4.1 Local problem

Under A3 and A4, in the neighbourhood of a given position along the boundary, the oxide occupies a half-strip $0 \leq x \leq L$ with L , D_O^{eff} and the reaction law treated as constants, and the concentration satisfies the steady equation

$$\frac{\partial^2 C_O}{\partial x^2} + \frac{\partial^2 C_O}{\partial y^2} = 0, \quad \frac{\partial C_O}{\partial x} \Big|_{x=0} = 0, \quad -D_O^{\text{eff}} \frac{\partial C_O}{\partial x} \Big|_{x=L} = J_r(C_O(L, y)). \quad (14)$$

4.2 Separable solutions

Seek $C_O = X(x)Y(y)$. Substituting into the Laplace equation and dividing by XY ,

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\left(\frac{\mu}{L}\right)^2, \quad (15)$$

where the separation constant is written as $-(\mu/L)^2$ with $\mu > 0$, because the x -problem must be oscillatory (it has a homogeneous Neumann condition at $x = 0$ and a sink at $x = L$) and the y -dependence must decay into the boundary. Then

$$X'' + \left(\frac{\mu}{L}\right)^2 X = 0, \quad X'(0) = 0 \Rightarrow X = \cos\left(\frac{\mu x}{L}\right); \quad Y'' - \left(\frac{\mu}{L}\right)^2 Y = 0, \quad Y \rightarrow 0 \Rightarrow Y = e^{-\mu y/L}. \quad (16)$$

Hence

$$C_O(x, y) = A \cos\left(\frac{\mu x}{L}\right) e^{-\mu y/L}, \quad (17)$$

where A is an amplitude and μ is to be fixed by the interfacial condition. Different values of μ give different modes; the fundamental mode has the smallest μ and the slowest decay in y , and by A5 it is the one retained.

4.3 Averages and flux of the mode

From (17), with the common factor $e^{-\mu y/L}$ absorbed into A ,

$$\bar{C}_O = \frac{A}{L} \int_0^L \cos\left(\frac{\mu x}{L}\right) dx = A \frac{\sin \mu}{\mu}, \quad (18)$$

$$C_O^{\text{int}} = A \cos \mu, \quad (19)$$

$$C_O^c = A, \quad (20)$$

$$J_r = -D_O^{\text{eff}} \left. \frac{\partial C_O}{\partial x} \right|_{x=L} = \frac{D_O^{\text{eff}} A \mu \sin \mu}{L}. \quad (21)$$

Eliminating the amplitude with (18), $A = \mu \bar{C}_O / \sin \mu$, gives the two relations of (2):

$$J_r = \frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L}, \quad C_O^{\text{int}} = \bar{C}_O \mu \cot \mu, \quad C_O^c = \frac{\mu \bar{C}_O}{\sin \mu}. \quad (22)$$

Both J_r and C_O^{int} are expressed through $\bar{C}_O$ and μ only. Note that $\mu \cot \mu$ decreases monotonically from 1 at $\mu = 0$ to 0 at $\mu = \pi/2$, so $0 \leq C_O^{\text{int}} \leq \bar{C}_O$ for $\mu \in [0, \pi/2]$.

4.4 Consistency with the averaged equation

Substituting the mode into the steady form of (13) with constant L and D_O^{eff} ,

$$D_O^{\text{eff}} L \frac{\partial^2 \bar{C}_O}{\partial y^2} = D_O^{\text{eff}} L \left(\frac{\mu}{L}\right)^2 \bar{C}_O = \frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L} = J_r, \quad (23)$$

with no residual. The averaged equation and the mode closure are therefore mutually consistent; no additional approximation is introduced by combining them.

5 The equation for μ

5.1 General (nonlinear) reaction law

Imposing the interfacial condition of (14) on the mode, and using (22) to express both sides through $\bar{C}_O$ and μ ,

$$g(\mu) \equiv \frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L} - J_r(\bar{C}_O \mu \cot \mu) = 0. \quad (24)$$

This is Eq. (1). It is a single scalar equation in μ ; the inputs are the local values of $\bar{C}_O$, L , D_O^{eff} and the metal concentrations that enter $J_r(\cdot)$.

Existence and uniqueness. On $(0, \pi/2)$ the first term increases monotonically from 0; the second term decreases monotonically, because $J_r(\cdot)$ is nondecreasing and $\mu \cot \mu$ is decreasing. Hence g is strictly increasing, with

$$g(0^+) = -J_r(\bar{C}_O), \quad g(\pi/2) = \frac{\pi^2 D_O^{\text{eff}} \bar{C}_O}{4L} - J_r(0). \quad (25)$$

If $J_r(\bar{C}_O) > 0$ (oxidation possible at the local average concentration) and $J_r(0) \leq 0$ (no consumption at zero oxygen, which holds for all four rate laws, including the magnetite law below its threshold), there is exactly one root $\mu^* \in (0, \pi/2)$. If $J_r(\bar{C}_O) \leq 0$, no oxygen is consumed: set $\mu = 0$, $J_r = 0$, $C_O^{\text{int}} = \bar{C}_O$.

Newton iteration. The derivative is

$$g'(\mu) = \frac{2D_O^{\text{eff}} \mu \bar{C}_O}{L} - J_r'(C_O^{\text{int}}) \bar{C}_O \frac{d}{d\mu}(\mu \cot \mu), \quad \frac{d}{d\mu}(\mu \cot \mu) = \frac{\sin \mu \cos \mu - \mu}{\sin^2 \mu} < 0, \quad (26)$$

so $g' > 0$ throughout and Newton's method converges from any bracketed start. A safeguarded Newton (fall back to bisection if an iterate leaves $(0, \pi/2)$) handles the kink of the magnetite law at $C_O^{\text{int}} = C_O^{\text{th}}$. A good initial guess is the linearized value of Sec. 5.2.

Algorithm per grid point and time step.

1. Inputs: $\bar{C}_O$, L , D_O^{eff} , $C_{\text{Cr}}^{\text{GB}}$, $C_{\text{Fe}}^{\text{GB}}$, $C_{\text{Si}}^{\text{GB}}$.
2. If $J_r(\bar{C}_O) \leq 0$: $\mu = 0$, $J_r = 0$, $C_O^{\text{int}} = \bar{C}_O$. Stop.
3. Solve $g(\mu) = 0$ on $(0, \pi/2)$ by safeguarded Newton.
4. $J_r = D_O^{\text{eff}} \mu^2 \bar{C}_O / L$; $C_O^{\text{int}} = \bar{C}_O \mu \cot \mu$; the individual $J_r^k(C_O^{\text{int}})$ give the metal sinks, Eqs. (eq:sink-cr)–(eq:sink-si) of `main.tex`, and the growth of each L_k .
5. Use J_r as the sink in (13).

This replaces the present quasi-steady closure, Eq. (eq:qss-closure), which also solves one scalar equation per point; the unknown is now μ instead of C_O^{int} .

5.2 Linearized reaction law

For $J_r = k C_O^{\text{int}}$, Eq. (24) becomes

$$\frac{D_O^{\text{eff}} \mu^2}{L} = k \mu \cot \mu \quad \Longleftrightarrow \quad \mu \tan \mu = \frac{kL}{D_O^{\text{eff}}} \equiv \text{Da}, \quad (27)$$

the classical eigenvalue condition for a Robin boundary. Da is the ratio of the lateral diffusion resistance L/D_O^{eff} to the reaction resistance $1/k$. The concentration $\bar{C}_O$ drops out: μ depends only on Da.

$\text{Da} = kL/D_O^{\text{eff}}$	μ	$C_O^{\text{int}}/\bar{C}_O = \mu \cot \mu$	$J_r L/(D_O^{\text{eff}} \bar{C}_O) = \mu^2$
0	0	1	0
0.1	0.311	0.968	0.097
1	0.860	0.740	0.740
10	1.429	0.204	2.042
∞	$\pi/2$	0	$\pi^2/4 \approx 2.467$

Table 1: Fundamental mode for a linear reaction law.

5.3 Limits

- **Reaction-limited**, $\text{Da} \ll 1$: $\mu^2 \simeq \text{Da}$, so $J_r \simeq k \bar{C}_O$ and $C_O^{\text{int}} \simeq \bar{C}_O$. The interface sees the average concentration; the lateral profile is nearly uniform, with a parabolic deviation of relative size $O(\text{Da})$. The decay length along the boundary is $L/\mu \simeq \sqrt{D_O^{\text{eff}} L/k} \gg L$.
- **Diffusion-limited**, $\text{Da} \gg 1$: $\mu \rightarrow \pi/2$, so $J_r \rightarrow \pi^2 D_O^{\text{eff}} \bar{C}_O/(4L)$ and $C_O^{\text{int}} \rightarrow 0$. Oxygen is consumed as soon as it reaches the interface; the lateral profile is a quarter cosine; the decay length along the boundary is $2L/\pi \approx 0.64L$.

Between these limits μ interpolates smoothly. The closure never needs a shape factor: if one insists on writing $J_r = f D_O^{\text{eff}} (\bar{C}_O - C_O^{\text{int}})/L$, then $f = \mu^2/(1 - \mu \cot \mu)$, which runs from 3 to $\pi^2/4$, but f plays no role in the algorithm.

6 Range of validity

1. **Higher modes.** The interfacial condition admits a sequence of eigenvalues $\mu_0 < \mu_1 < \dots$, with $\mu_n \in (n\pi, n\pi + \pi/2)$ for a Robin condition. Each decays as $e^{-\mu_n y/L}$. The first higher mode decays at least as fast as $e^{-\pi y/L}$, i.e. over a distance $\lesssim L/\pi \approx 0.3L$, compared with $\geq 2L/\pi$ for the fundamental. Higher modes are excited where the boundary data change abruptly: the surface entry, the oxidation tip, and any position where L , D_O^{eff} or the metal concentrations jump. Away from such places, only the fundamental survives (A5).

2. **Slowly varying geometry (A3).** The mode is derived for a strip of constant width. For a wedge-shaped oxide the closure is applied with the local $L(y)$; the error is $O(\partial L/\partial y)$, the same order as the terms dropped in (11).
3. **Quasi-steady lateral profile (A4).** Requires L^2/D_O^{eff} to be short compared with the time scales of $\bar{C}_O(t)$ and $L(t)$. This can be checked from the simulation output.
4. **Early stage.** When the oxidized zone is as deep as it is wide, the decay length is comparable to L and the whole problem is two-dimensional. The one-dimensional description is meaningful only once the oxidized zone is slender; the initial aspect-ratio-one stage lies outside its scope.
5. **Uniform D_O^{eff} across the section (A1).** This is what allows the same D_O^{eff} in (13) and (14). It follows from the local mixing of phases assumed in `main.tex`.

7 Summary of the model change

Current formulation in `main.tex`:

$$\frac{\partial C_O}{\partial t} = \frac{\partial}{\partial y} \left(D_O^{\text{eff}} \frac{\partial C_O}{\partial y} \right) - \frac{J_r}{\delta_{\text{GB}}}, \quad J_r = D_O^{\text{eff}} \frac{C_O - C_O^{\text{int}}}{\sum_k L_k} = \sum_k J_r^k(C_O^{\text{int}}). \quad (28)$$

Proposed formulation (cross-section average over the whole oxide, fundamental-mode closure):

$$\frac{\partial}{\partial t}(L \bar{C}_O) = \frac{\partial}{\partial y} \left(D_O^{\text{eff}} L \frac{\partial \bar{C}_O}{\partial y} \right) - J_r, \quad J_r = \frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L}, \quad \frac{D_O^{\text{eff}} \mu^2 \bar{C}_O}{L} = \sum_k J_r^k(\bar{C}_O \mu \cot \mu). \quad (29)$$

What changes: (i) the storage width and along-boundary conductance are the time-dependent half-width L instead of the fixed δ_{GB} , with L inside the time derivative; (ii) the lateral closure is the fundamental Laplace mode, valid from the reaction-limited to the diffusion-limited regime, with μ obtained pointwise by Newton's method; (iii) L is the mid-plane-to-interface distance. What does not change: the reaction laws, the metal sinks, the definition of D_O^{eff} as a local volume-fraction average, and the structure of the pointwise scalar solve.